

Part I

Recitation Questions

Problem 1

February 22, 2026

Problem 1

Problem 1 Compute the following indefinite integrals.

(a) $\int 2t \sin(t^2) \, dt.$

(b) $\int \cos(x) \sqrt{\sin(x)} \, dx.$

(c) $\int \left(3t^2 - 4 + \frac{1}{t} \right) e^{t^3 - 4t + \ln(t) - 9} \, dt.$

Problem 2

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Problem 2

Problem 2 Evaluate the following definite integrals:

(a) $\int_{-2}^1 t^2 \sin(t^3) \, dt$

(b) $\int_0^{1/2} \frac{13e^u}{3e^u - 5} \, du$

(c) $\int_0^{\pi/2} \frac{\cos(x)}{4 + \sin(x)} \, dx$

(d) $\int_1^4 \frac{e^{\sqrt{x}}}{3\sqrt{x}} \, dx$

(e) $\int_{\pi/3}^{\pi/2} \sin(x) \sec^2(\cos(x)) \, dx$

Problem 3

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Problem 3

Problem 3 Suppose that $\int_1^3 f(x) dx = 4$

(a) Evaluate the following integrals.

(i) $\int_0^2 f(x+1) dx$

(ii) $\int_1^9 3 \frac{f(\sqrt{x})}{\sqrt{x}} dx$

(iii) $\int_{\frac{1}{3}}^1 f(3x) dx$

(iv) $\int_2^4 3f(x-1) dx$

(v) $\int_0^{\sqrt{2}} 3xf(x^2+1) dx$

(b) Assume that f is odd. Evaluate $\int_{-3}^{-1} f(x) dx$

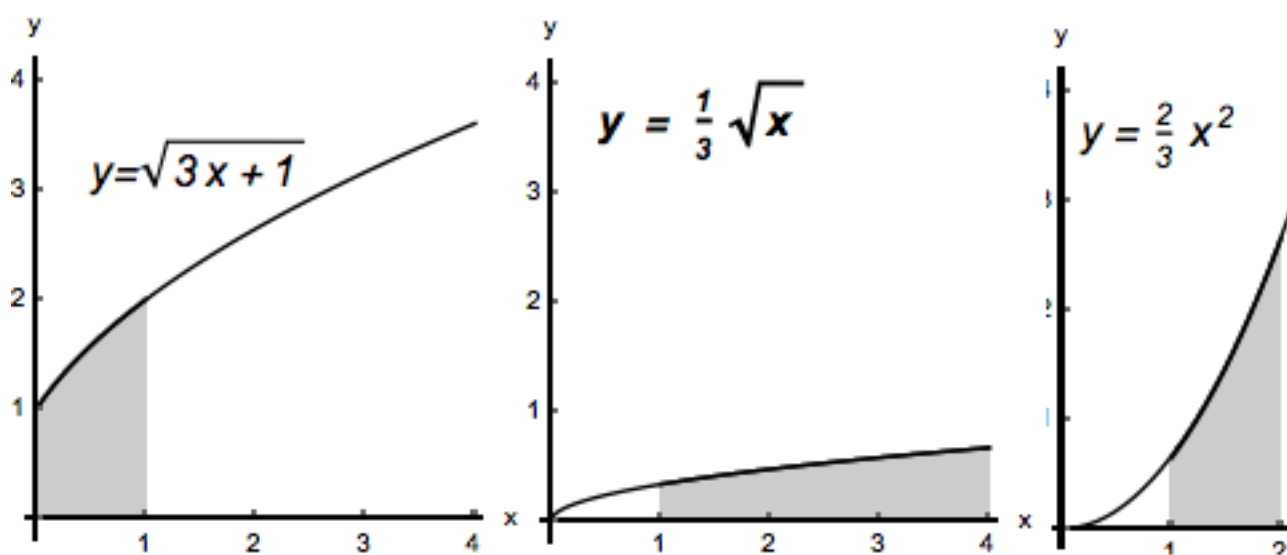
(c) Assume that f is even. Evaluate $\int_{-3}^{-1} f(x) dx$

Problem 4

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Problem 4

Problem 4 Show that the areas of the three shaded regions in the figure below are equal. Find this area.



Problem 5

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Problem 5 Find the error in the following “solution”:

Find $\int_{-2}^2 \frac{1}{x^8 - 1} dx$.

$$\text{let } u = x^4$$

$$\text{then } du = 4x^3 dx$$

$$\frac{du}{4x^3} = dx$$

$$\text{and } u^{1/4} = x$$

$$u(2) = 2^4 = 16$$

$$u(-2) = (-2)^4 = 16$$

$$= \int_{16}^{16} \frac{1}{(u^2 - 1)} \cdot \frac{du}{4u^{3/4}}$$

but an integral $\int_a^a f(x) dx = 0$

$$\text{so } \int_{-2}^2 \frac{1}{x^8 - 1} dx = 0$$

Problem 6

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Problem 6

Problem 6 Find the error in the following “solution”, correct the error (or errors) and evaluate the integral.

Find $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x) \sqrt{1 - \cos^2(x)} dx$.

“Solution”:

Since $\sin(x) = \sqrt{1 - \cos^2(x)}$, let $u = \sin(x)$. Then $du = \cos(x)dx$;

when $x = -\frac{\pi}{2}$, $u = \sin\left(-\frac{\pi}{2}\right) = -1$; when $x = \frac{\pi}{2}$, $u = \sin\left(\frac{\pi}{2}\right) = 1$. Therefore

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x) \sqrt{1 - \cos^2(x)} dx = \int_{-1}^1 u \cdot du = 0,$$

due to symmetry (integrating an odd function over a symmetric interval, $[-1, 1]$).

Problem 7

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Problem 7

Problem 7 Compute the integral: $\int \frac{1+3x}{4+4x^2} dx$.

Problem 8

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Problem 8

Problem 8 Evaluate the integral $\int \frac{x^2}{1+x^2} dx$.

Problem 9

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Problem 9

Problem 9 Compute the integral: $\int \sec^2(x) \tan(x) \, dx$.

Problem 10

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Problem 10

Problem 10 *What are two substitutions that can be used to evaluate the integral*

$$\int x\sqrt{x+8} \, dx$$

Problem 11

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Problem 11

Problem 11 Compute the integral: $\int \frac{x}{\sqrt{x-4}} dx$.

Problem 12

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Problem 12 Evaluate the following indefinite integrals:

(a) $\int \frac{13x^7}{\sqrt{3x^4 - 5}} dx$

(b) $\int \frac{u^3}{u^2 - 3} du$

(c) $\int e^{t^2 + \ln(t)} dt$